The Potential Outcome Framework

PS200C, Causal Inference, Spring 2024

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What is causality?

What do we mean when we say X causes Y, or Y was caused by X?

Thoughts?

Reading: Chapers 1 and 2 from “What If" (Hernan & Robins) (google it)

Flavors of causality

Effect of causes. One kind of causal question the most common type of causal question in applied areas-regards the *effect* of some potential cause, e.g. what is the effect of:

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Other/more general/complicated concepts can be embedded in a deep enough framework (namely structural causal models)

Two languages

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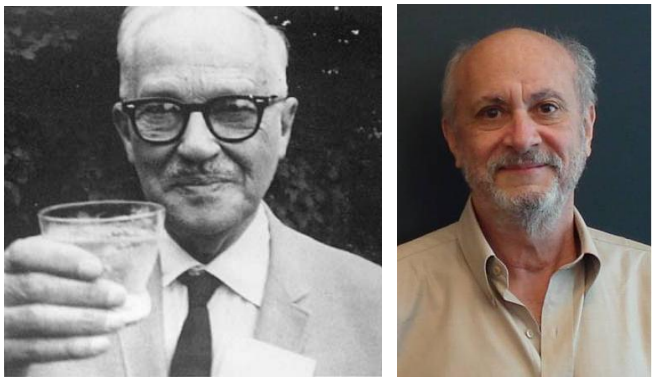

Figure: Neyman and Rubin

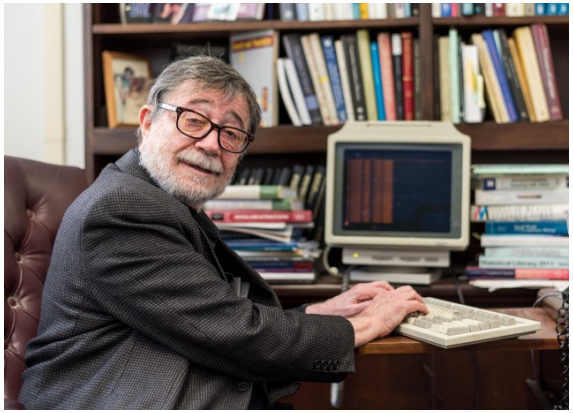

Figure: Judea Pearl

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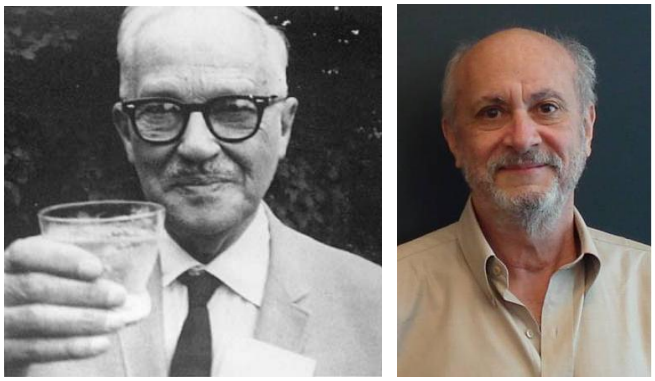

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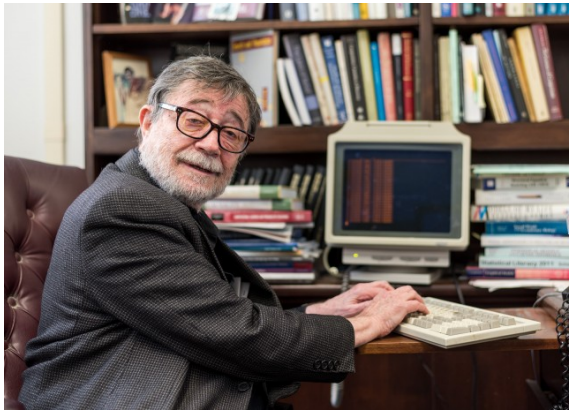

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We will use both extensively, but start with potential outcomes.

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- The trouble is you do not get to see both Y_{1i} and Y_{0i} for unit i. This is the fundamental problem of causal inference.
- Everything we do will be about filling in the "missing" potential outcome for each unit through various **assumptions**. That is it!

Fundamental Quantities

- Y_{1i} and Y_{0i}, the potential outcomes
- D_i, the treatment status of unit i
- Observed outcome, $Y_i = D_i Y_{1i} + (1 - D_i) Y_{0i}$
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Quantities of interest

Using these, we will soon be interested in things like:

- Average treatment effect (ATE): $\mathbb{E}[Y_{1i} - Y_{0i}] = \mathbb{E}[\tau]$
- Average treatment effect among treated (ATET)

$$\mathbb{E}[\tau_i | D_i = 1] = \mathbb{E}[Y_{1i} - Y_{0i} | D_i = 1]$$

- Average treatment effect among controls (ATEC)

$$\mathbb{E}[\tau_i | D_i = 0] = \mathbb{E}[Y_{1i} - Y_{0i} | D_i = 0]$$

Warning: regression (or comparison-based) goggles

- You are used to thinking D may be associated with Y, especially so if D affects Y.
- The *potential* outcomes are not the outcomes: they don't change when you change treatment, they say what would happen under a given treatment.
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If I haven't made it clear: we don't really care about group comparisons, we care about within-unit counterfactual comparisons.

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- one-version-of-treatment – d means the same thing all the time...but we can almost always define our way around this so it is not much of a problem.

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We can relax these in various ways, including:

- "treatment variation irrelevance" (VanderWeele 2009) or redefining treatment
- expanding POs to index treatment of others in some limited way

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- "Changing Trump to female" would change countless subsequent experiences (and the probability of being in the sample of presidents anyway).

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Later: you will see that our intuitions about potential outcomes and how they relate to observed variables can go wrong once you start conditioning on things (including sample selection)...graphical approaches will help us see that.

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- Can you say anything about expected direction of bias?

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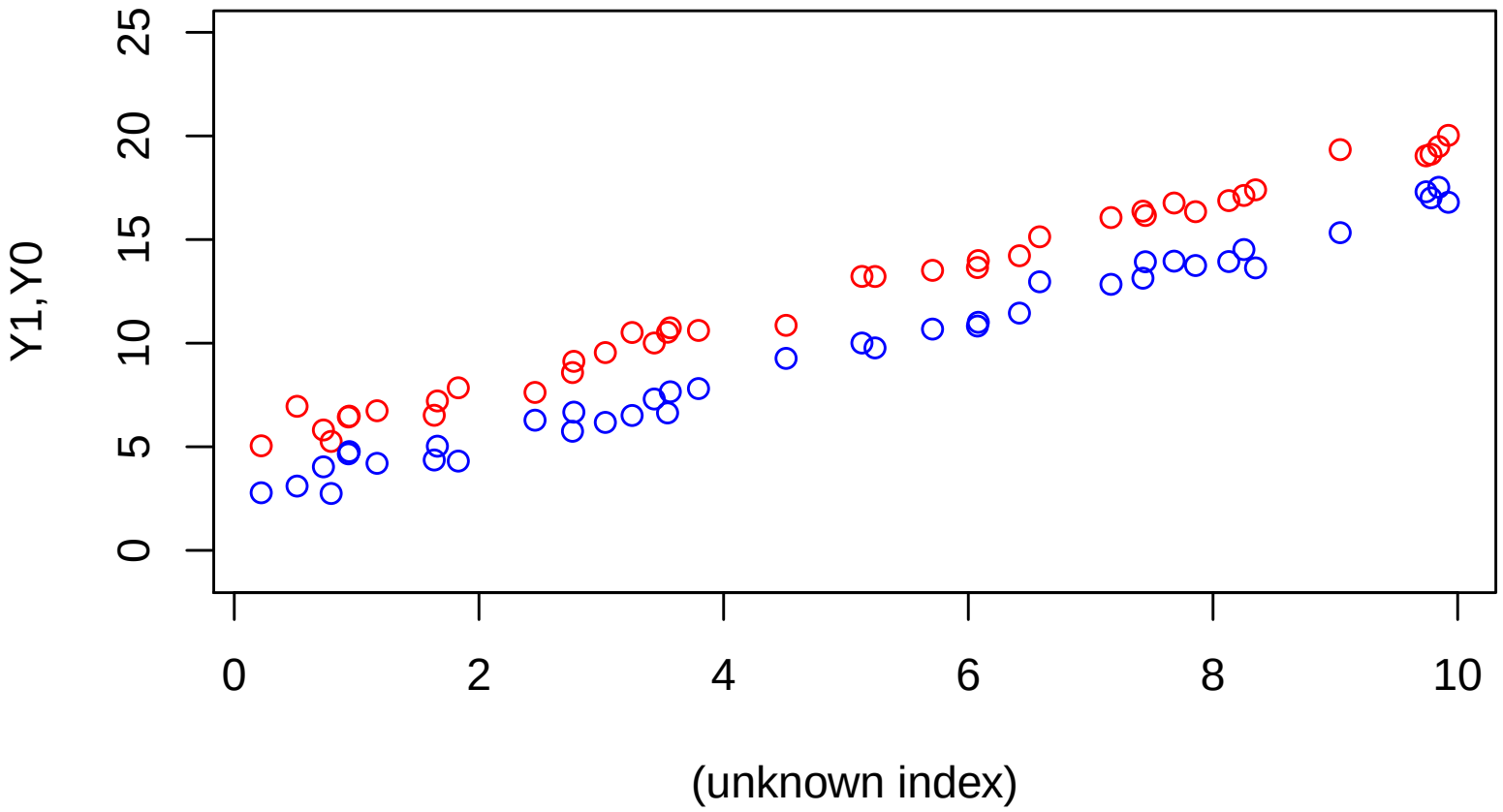
Which of these scenarios are we usually in with observational data?

Visual Practice

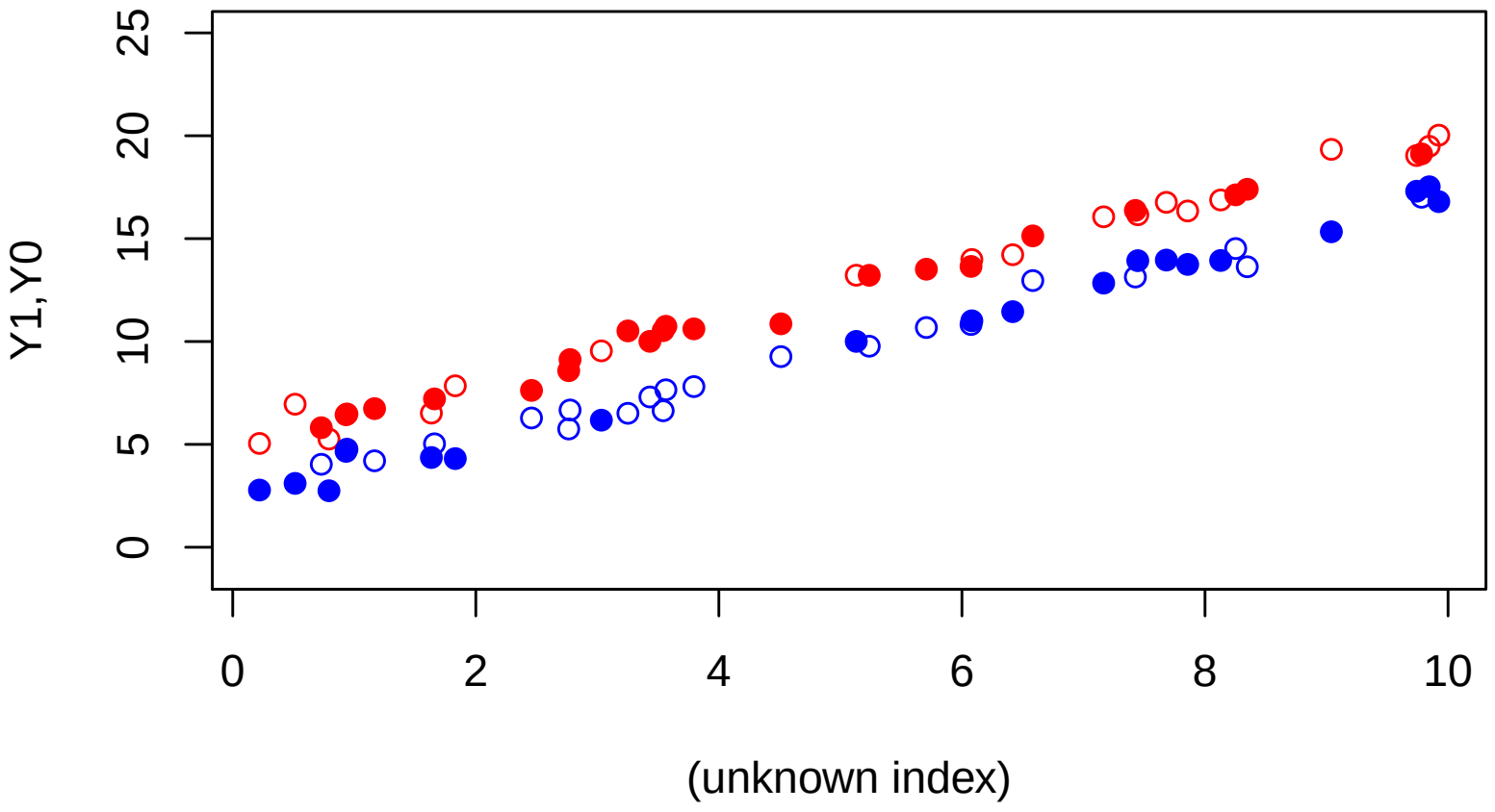
Let's looks at Y_{1i} and Y_{0i} alone first.



True $\mathbb{E}[Y_{1i} - Y_{0i}] = ATE = 3$

Random Assignment of Treatment

Now suppose $D_i = 1$ is randomly assigned. We see only the filled dots:



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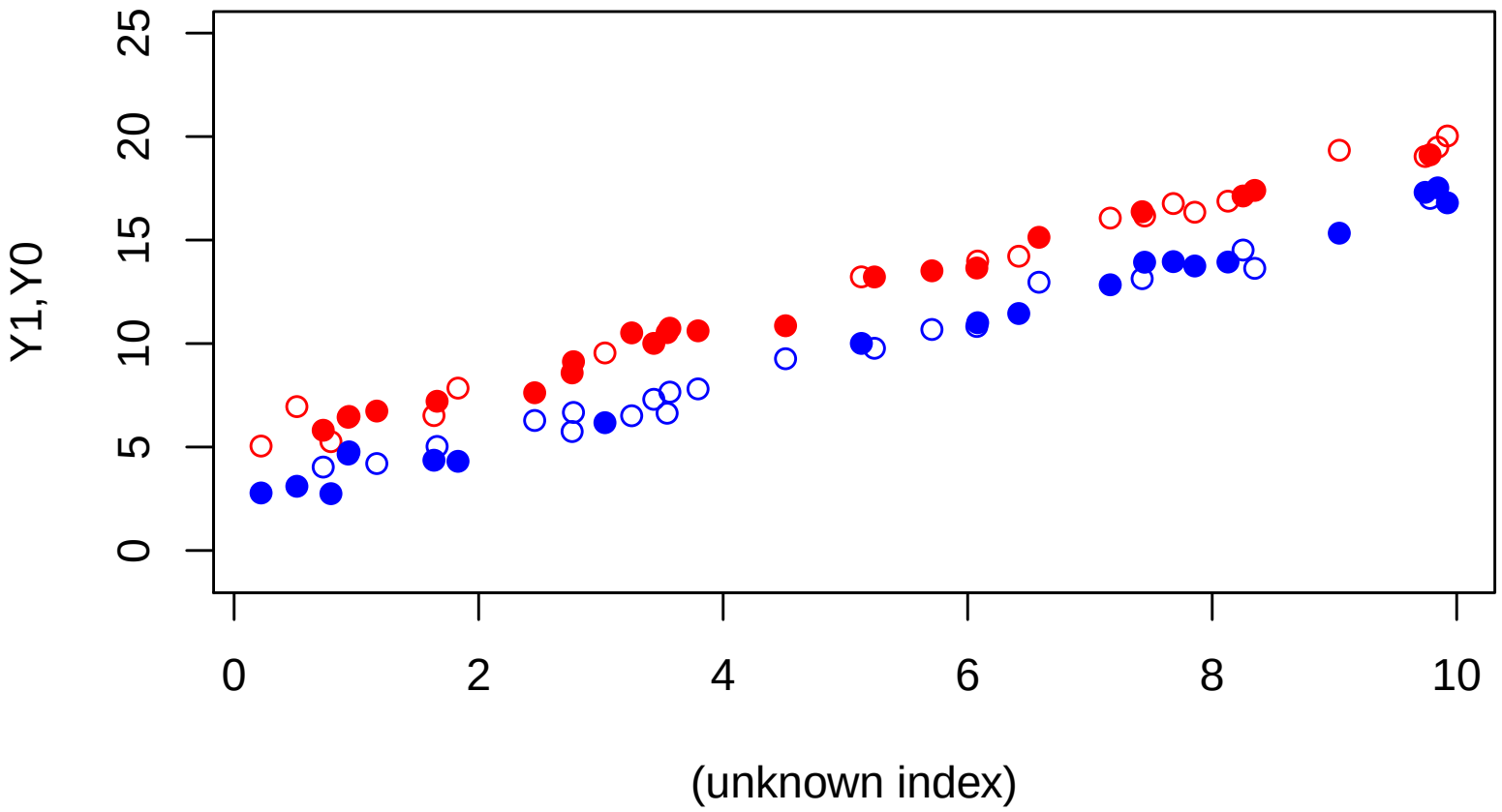
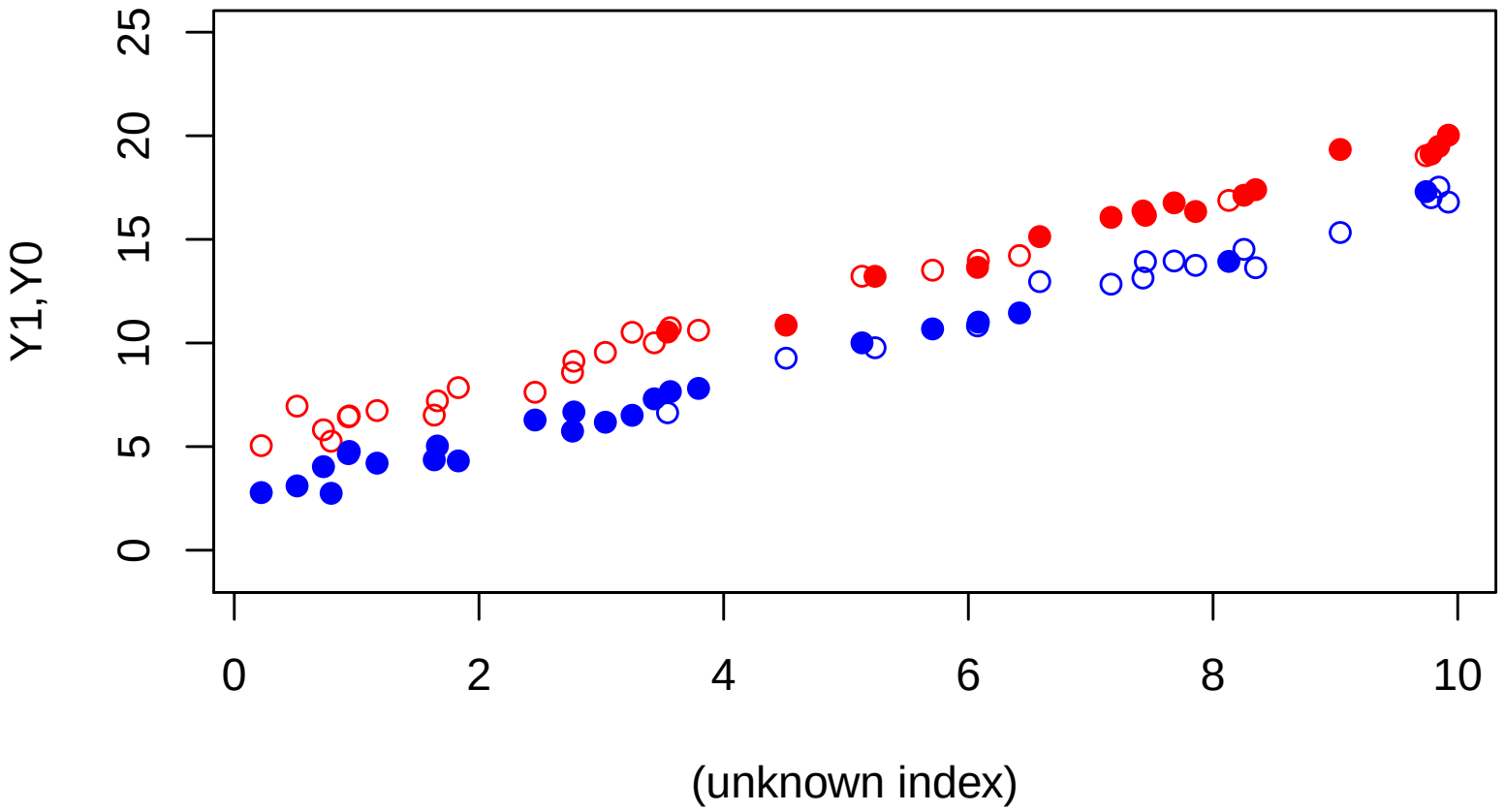


We get

$$\hat{\mathbb{E}}[Y_i|D_i = 1] - \hat{\mathbb{E}}[Y_i|D_i = 0] = \widehat{ATE} = 3.01$$

Non-Random Assignment of Treatment 1

Suppose $Pr(D_i = 1)$ increases to the right. Now we observe (filled dots):



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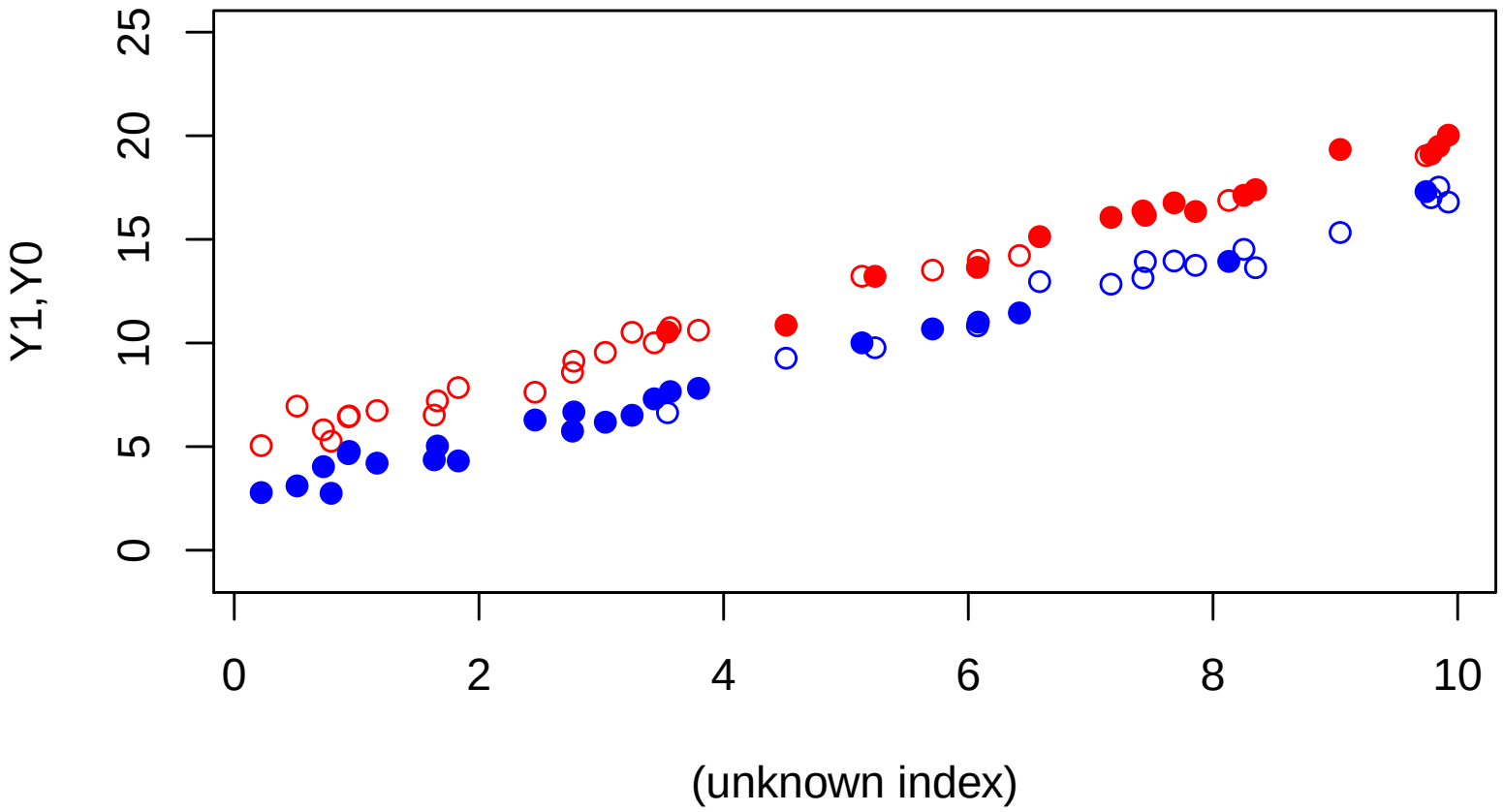

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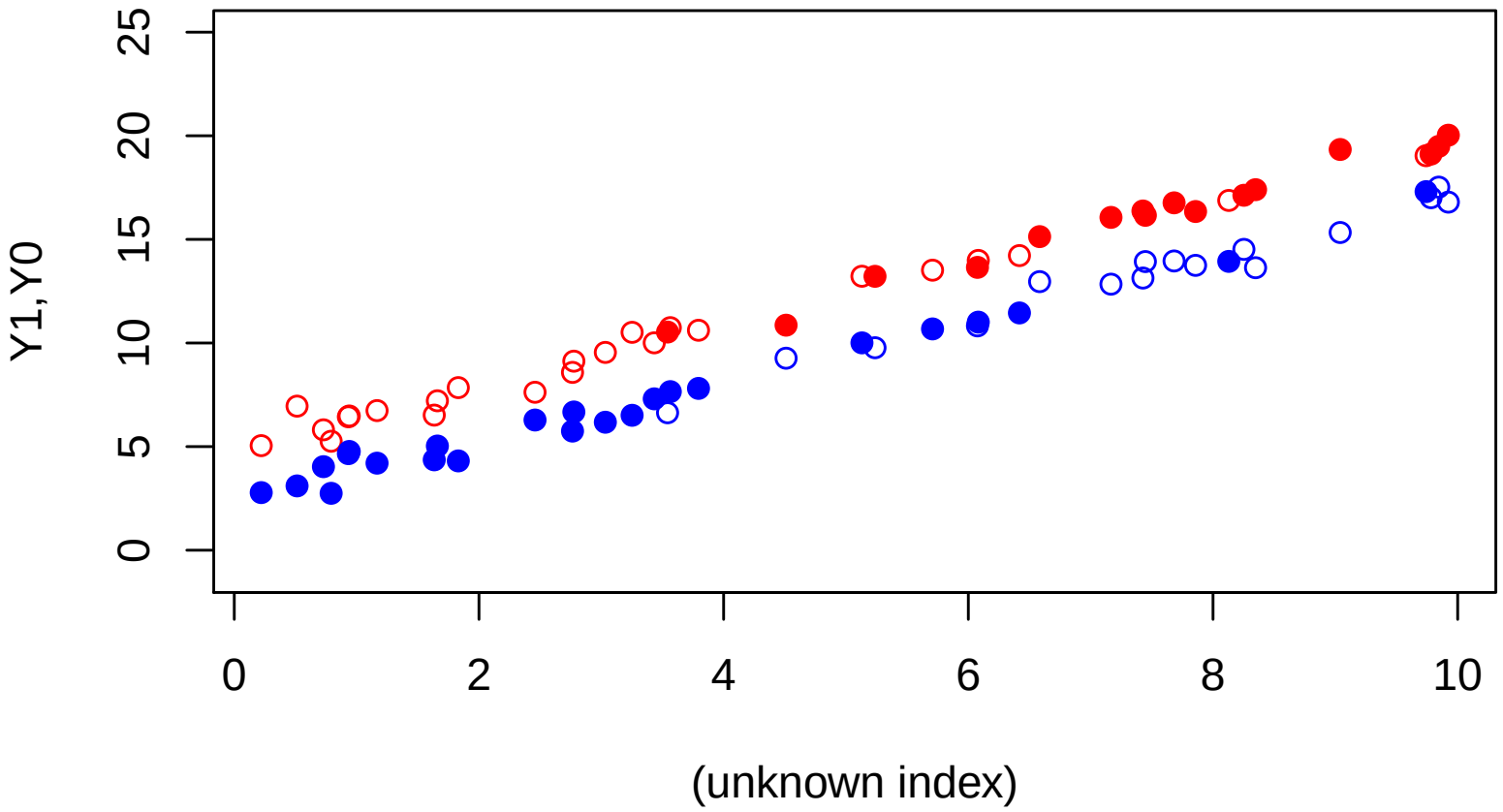

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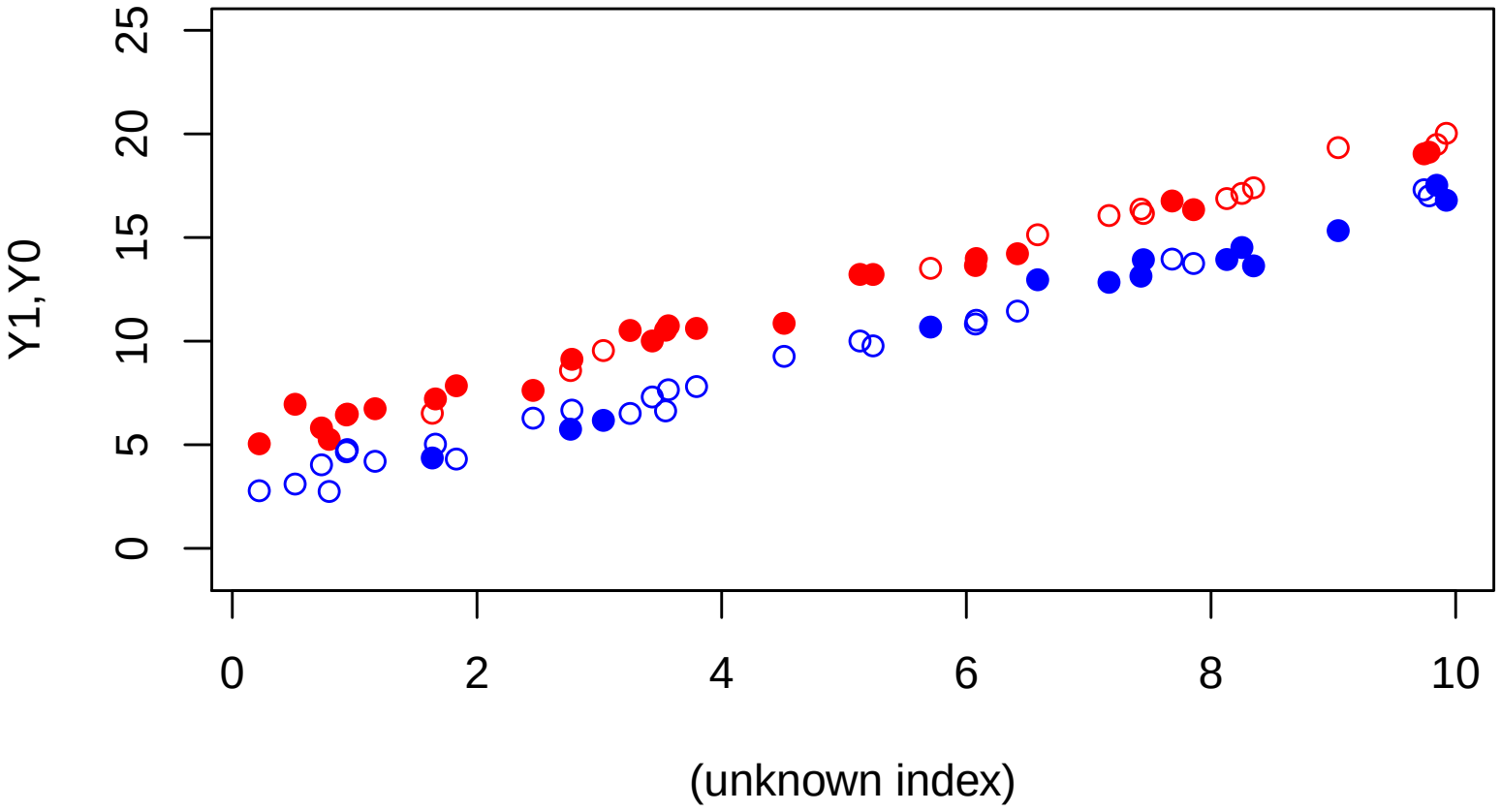
$$\hat{\mathbb{E}}[Y_i|D_i = 1] - \hat{\mathbb{E}}[Y_i|D_i = 0] = \widehat{ATE} = 9.97$$

Non-Random Assignment of Treatment 2

Now suppose $Pr(D_i = 1)$ decreases to the right. We observe (filled dots):



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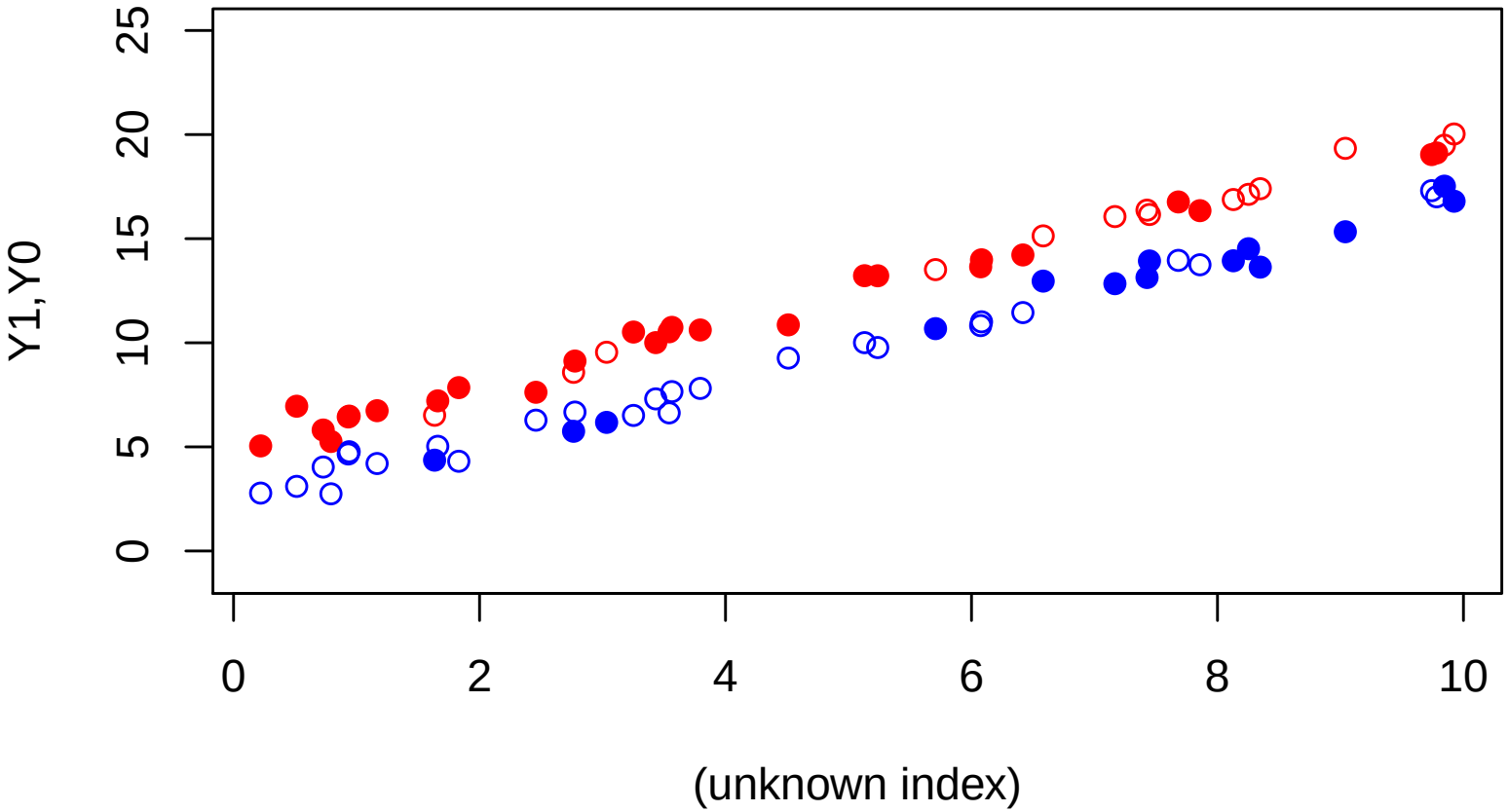

What difference in means do you expect now?

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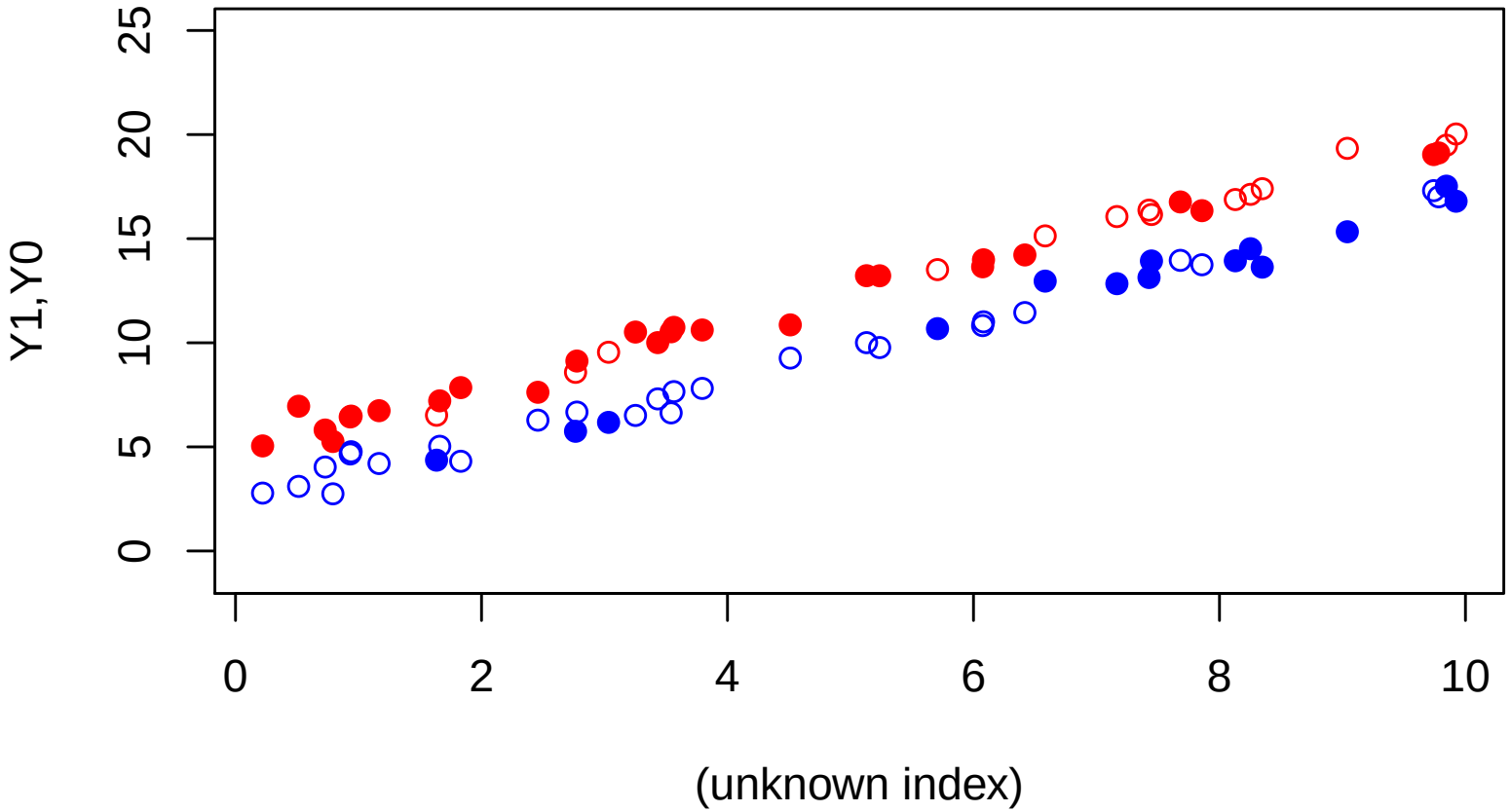

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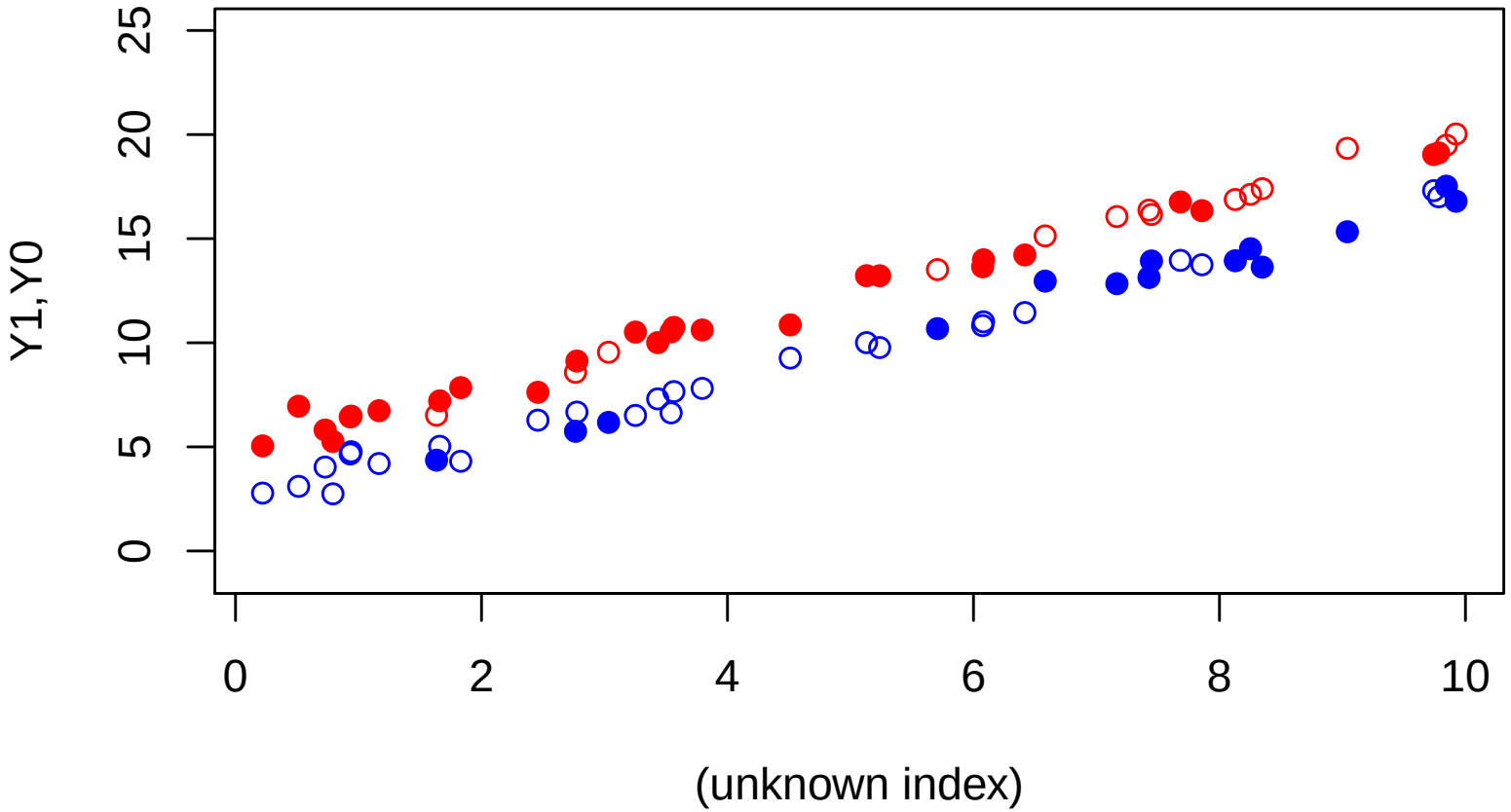

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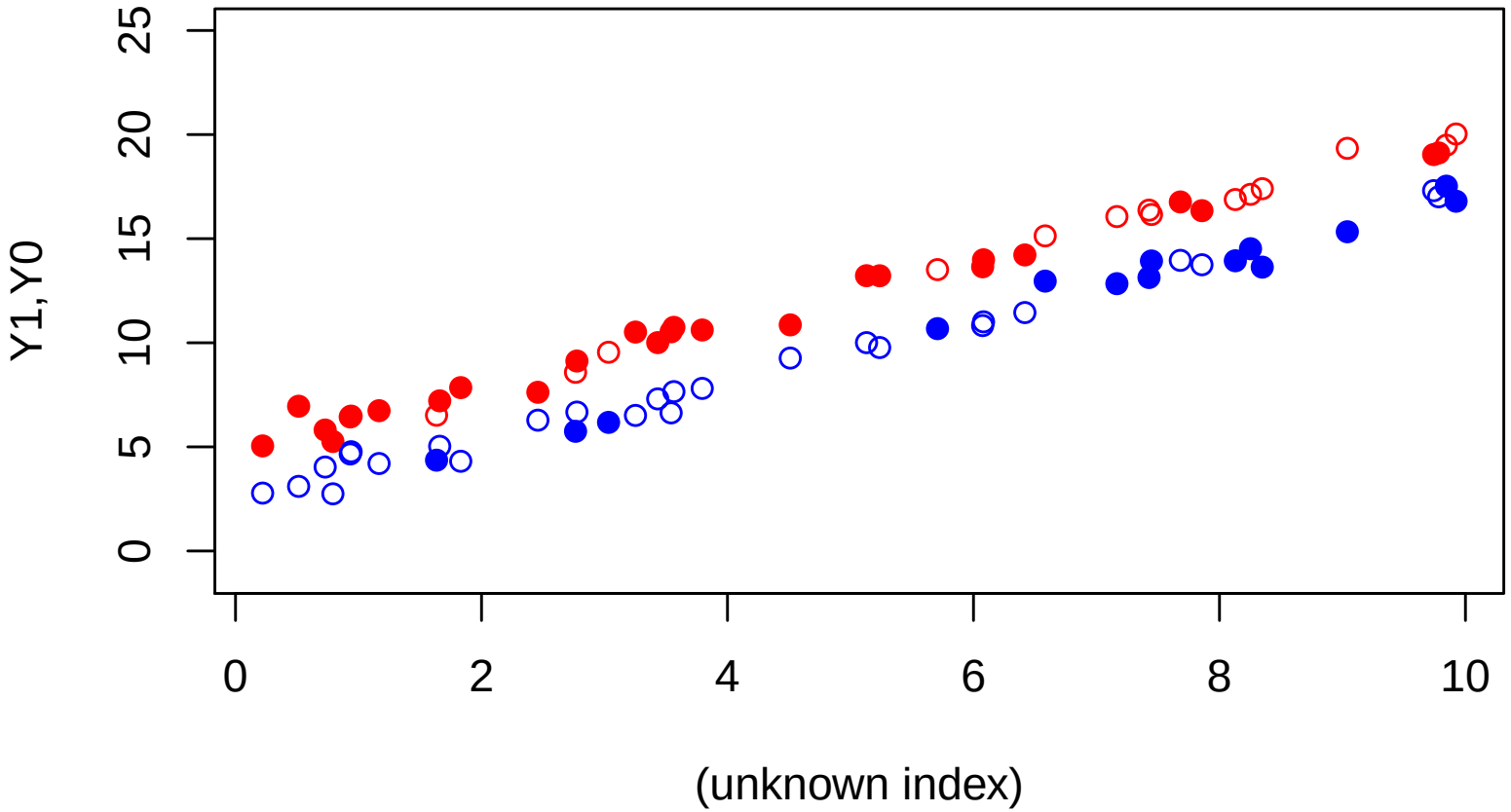

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What can we do about this? Nothing, yet

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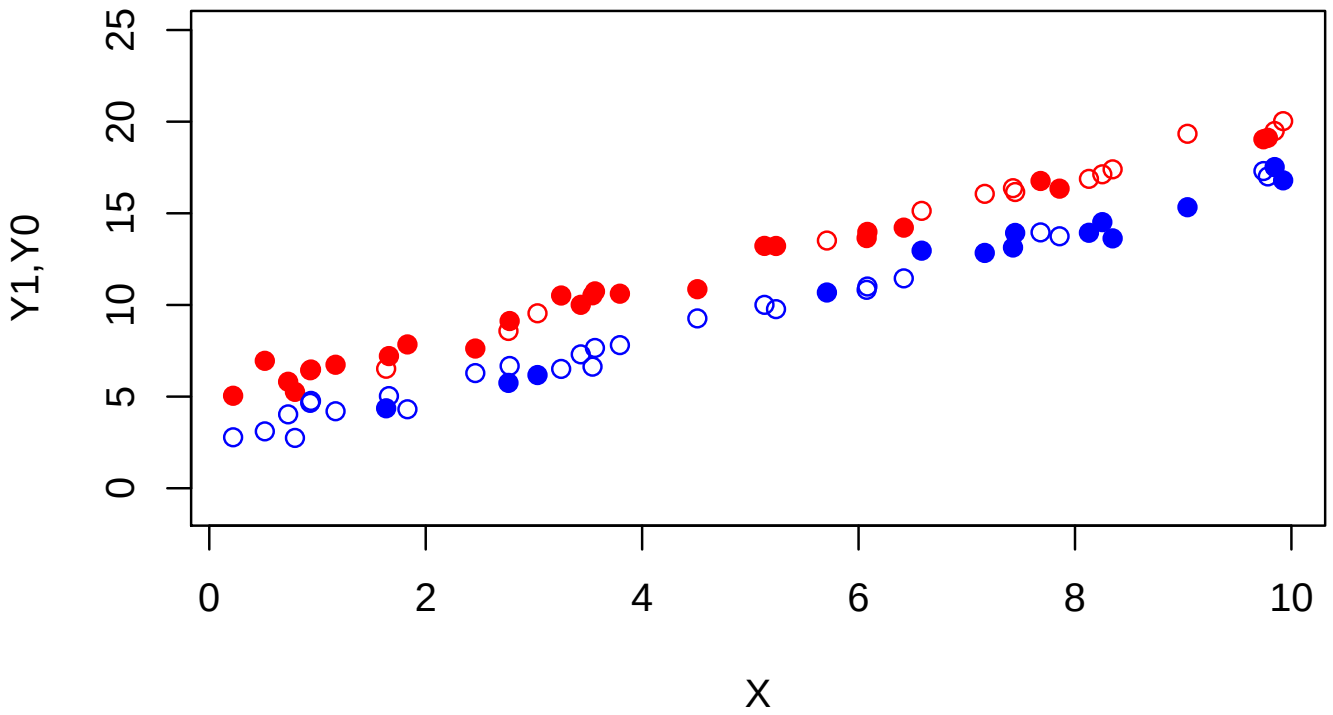

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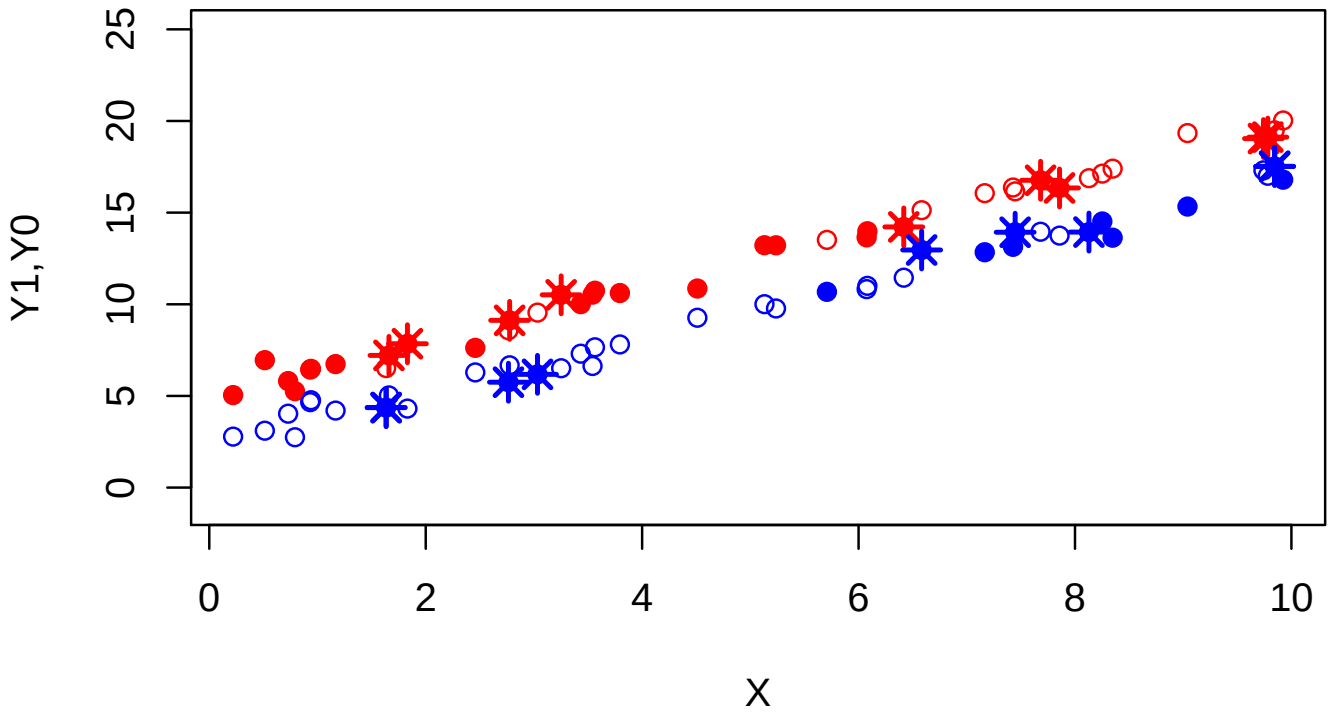

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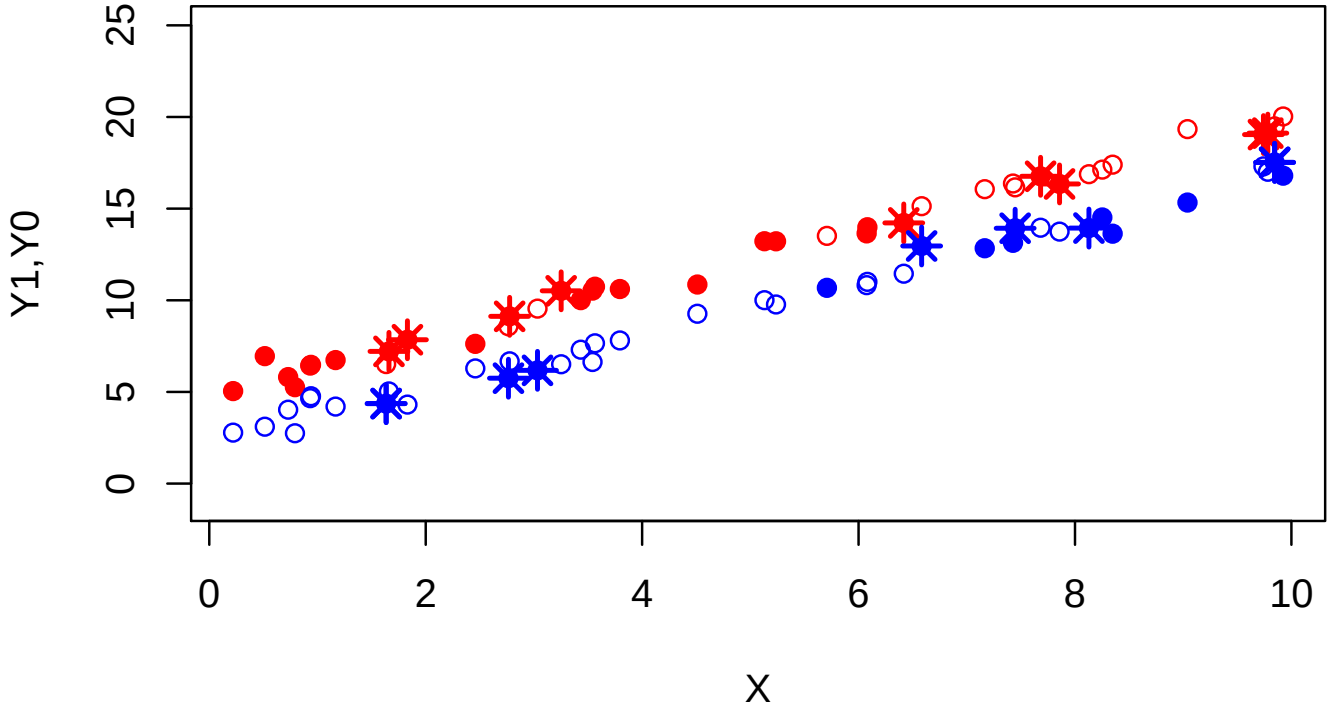

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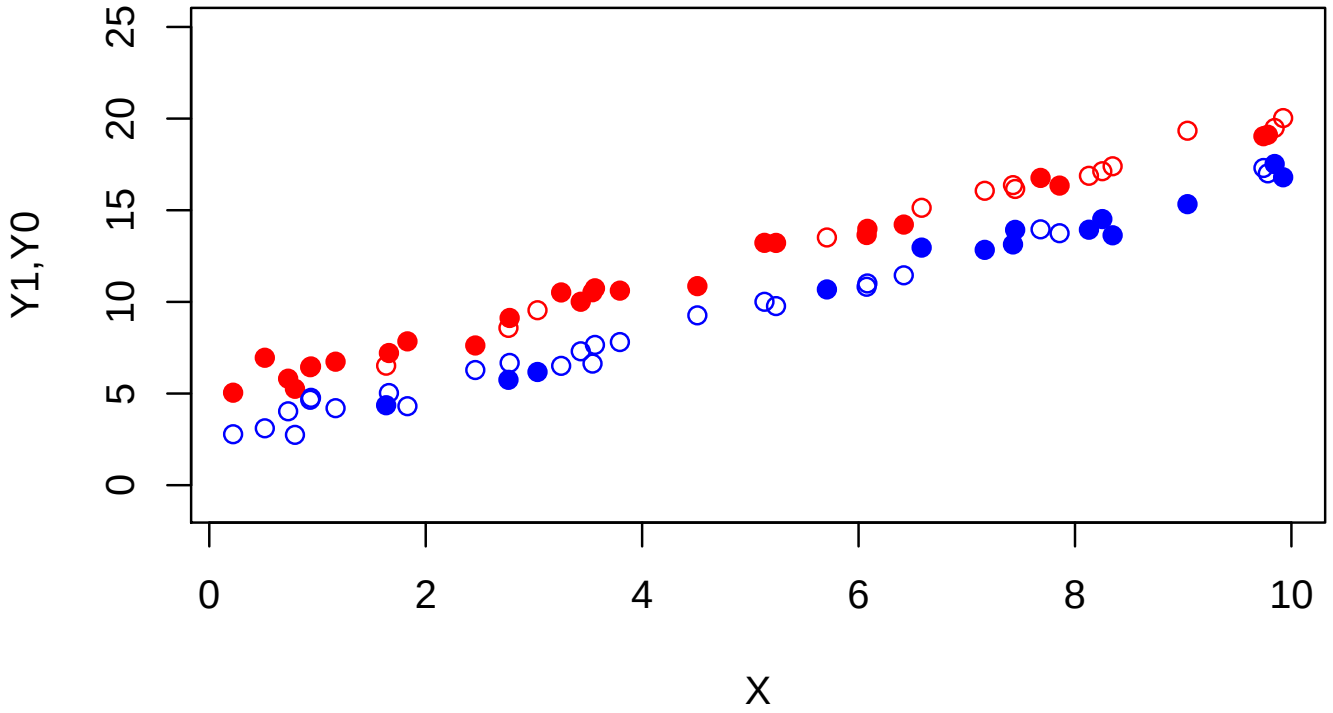
Practice

Looking at this plot, describe the meaning of the ATE, ATT, ATC, & $ATE(X)$ graphically



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Causal inference in the presence of interference is an area of active research. e.g. sometimes "spillover" or "saturation" is deliberately varied

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Imagine a study population with 4 units:

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What are the terms in blue & green?

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Example: Human rights treaties

- Maybe countries willing to sign a human rights treaty have better human rights (absent any effect of the treaty itself): bias $= \mathbb{E}[Y_{0i}|D_i=1]-\mathbb{E}[Y_{0i}|D_i=0] \neq 0$

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Those represent important assignment mechanisms with hallowed names in varied fields:

- random assignment, ignorability, or exchangeability
- selection on observables, conditional ignorability
- bad news

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From here: random assignment, then random-conditional-on-observables.